

Identifying errors in proofs - Set 2 - Solutions

Specification Err1, Err2.

Answer key

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Worked solutions

Question 1 ERR-07 · Identifying errors in proofs · spec Err2

A student writes the following argument.

I $0.1 > 0.01$.

II Taking logarithms to base 10 of both sides: $-1 > -2$.

III Multiplying both sides by $\log_{10} \frac{1}{2}$, which is negative, reverses the inequality:
 $-\log_{10} \frac{1}{2} < -2\log_{10} \frac{1}{2}$.

Which of the following is correct?

Answer: A

Logarithms are increasing functions, so line II is valid; $\log_{10} \frac{1}{2} < 0$, so reversing the inequality in line III is correct.

Full method - how and why

Sometimes the argument is simply valid. Check each step against the rules rather than assuming a trap.

1 Line I: $0.1 > 0.01$ ✓.

why: $\frac{1}{10} > \frac{1}{100}$.

2 Line II: $\log_{10}$ is an **increasing** function, so it preserves the inequality: $\log_{10} 0.1 = -1$ and $\log_{10} 0.01 = -2$, and indeed $-1 > -2$ ✓.

why: Applying an increasing function to both sides keeps the direction; $\log_{10} 10^{-1} = -1$.

3 Line III: $\log_{10} \frac{1}{2} < 0$ because $\frac{1}{2} < 1$. Multiplying an inequality by a negative number reverses it, which is exactly what is written ✓.

why: $-1 > -2$ becomes $-1 \times c < -2 \times c$ for $c < 0$.

4 So all three lines are correct.

why: The question is testing whether you can certify a valid argument, not only spot faults.

Where it goes wrong

- Assuming logarithms reverse inequalities - they do not; **log** is increasing on its domain.
- Thinking $\log_{10} \frac{1}{2}$ is positive; logs of numbers below 1 are negative.
- Misreading $\log_{10} 0.1$ as 1 rather than -1 .
- Hunting for an error simply because the question implies one might exist.

Takeaway Increasing functions preserve inequalities; multiplying by a negative reverses them. $\log_b x < 0$ exactly when $0 < x < 1$ (for $b > 1$).

Question 2 ERR-08 · Identifying errors in proofs · spec Err1

Claim: if n is an integer and n^2 is divisible by 4, then n is divisible by 4.

- I Suppose n^2 is divisible by 4.
- II Then $n^2 = 4k$ for some integer k .
- III So $n = 2\sqrt{k}$, and since n is an integer $\sqrt{k}$ is an integer.
- IV Hence n is even, so $n = 2m$, and $n^2 = 4m^2$ is divisible by 4, so n is divisible by 4.

Which of the following is true?

Answer: A

$n = 2$: $n^2 = 4$ is divisible by 4 but n is not. Lines I-III are correct deductions (n even). Line IV concludes ' n divisible by 4' from ' n^2 divisible by 4', which is exactly the unproved claim - circular and false.

Full method - how and why

A circular argument: the last line assumes the very thing being proved. Also, the claim is false.

- 1 Test the claim: $n = 2$ gives $n^2 = 4$, divisible by 4, but 2 is not. **Claim false.**

why: Establish this first; an error must exist.

- 2 Lines I-III: from $n^2 = 4k$, n^2 is even so n is even. The algebra $n = 2\sqrt{k}$ with $\sqrt{k}$ an integer is correct given n is an integer. Valid.

why: These lines legitimately establish that n is even.

- 3 Line IV: it writes $n = 2m$, computes $n^2 = 4m^2$ (true but irrelevant), and then concludes ' n is divisible by 4'. That conclusion does not follow from anything. **First error.**

why: Showing n^2 is divisible by 4 was the **hypothesis**; restating it is not progress. Divisibility of n by 4 would require m even, which is never shown.

- 4 Compare with the working proof for $8 \mid n^2 \Rightarrow 4 \mid n$: there the extra factor forces m even. Here there is no spare factor.

why: Seeing where the analogous proof gets its leverage shows exactly what is missing.

Where it goes wrong

- Accepting line IV because every individual equation in it is true - the **inference** is what fails.
- Blaming line III; introducing $\sqrt{k}$ is unusual but not invalid here.
- Accepting the claim because $4 \mid n \Rightarrow 4 \mid n^2$ (the converse) is true.

Takeaway A step that restates the hypothesis and then asserts the conclusion is circular. $4 \mid n^2$ gives only $2 \mid n$.

Question 3 ERR-11 · Identifying errors in proofs · spec Err1

Claim: $\int_{-1}^1 \frac{1}{x^2} dx = -2$.

- I An antiderivative of x^{-2} is $-x^{-1}$.
- II So the integral equals $\left[-\frac{1}{x}\right]_{-1}^1 = -1 - 1 = -2$.
- III Hence the value is -2 .

Which of the following is correct?

Answer: A

The integrand is positive everywhere it is defined, so a negative answer is a warning sign; the function is not defined on the whole interval and the integral does not exist in the ordinary sense.

Full method - how and why

The fundamental theorem of calculus needs the integrand to be defined (and continuous) throughout the interval.

- 1 Spot the warning sign: $\frac{1}{x^2} > 0$ wherever it is defined, yet the 'answer' is negative.
why: An integral of a positive function over an interval cannot be negative - the sign alone signals an error.
- 2 Line I is fine: $\frac{d}{dx}(-x^{-1}) = x^{-2}$ ✓.
why: Differentiate: $-(-1)x^{-2} = x^{-2}$.
- 3 Line II applies the fundamental theorem across $[-1, 1]$. But $\frac{1}{x^2}$ is undefined at $x = 0$, which lies **inside** the interval. **First error.**
why: The antiderivative is not continuous on the whole interval, so the theorem does not apply.
- 4 The arithmetic in line II is itself right ($-\frac{1}{1} - (-\frac{1}{-1}) = -1 - 1 = -2$); the fault is applying the theorem at all.
why: Correct arithmetic on an invalid application still gives nonsense.

Where it goes wrong

- Trying to 'fix' the sign to $+2$ - the integral does not exist in the ordinary sense, so no finite value is right.
- Blaming line I; the antiderivative is correct.
- Ignoring the discontinuity because the limits themselves are perfectly ordinary numbers.

Takeaway Before applying $\int_a^b f = F(b) - F(a)$, check f is defined on all of $[a, b]$; and a positive integrand can never give a negative integral.

Question 4 ERR-13 · Identifying errors in proofs · spec Err1

Claim: for all positive real x , $x + \frac{1}{x} \geq 2$.

I Suppose $x + \frac{1}{x} \geq 2$.

II Multiply by $x > 0$: $x^2 + 1 \geq 2x$.

III So $x^2 - 2x + 1 \geq 0$, i.e. $(x - 1)^2 \geq 0$, which is true.

IV Hence the claim is proved.

Which of the following best describes the argument?

Answer: A

Starting from the conclusion and reaching a truth does not prove the conclusion unless every step is reversible and the argument is presented in the forward direction. Here the steps are reversible, so the repair is to write it backwards.

Full method - how and why

Assuming the conclusion. The chain of steps is valid but it runs the wrong way.

1 Line I supposes $x + \frac{1}{x} \geq 2$ - which is exactly the claim to be proved. **First error.**

why: A proof may suppose the **hypothesis** ($x > 0$), never the conclusion.

2 Lines II and III are then correct algebra, ending at $(x - 1)^2 \geq 0$, a truth.

why: Deriving a true statement from an assumption proves nothing: false statements can also imply true ones.

3 The repair: run it backwards. Start from $(x - 1)^2 \geq 0$ (known true), expand to $x^2 - 2x + 1 \geq 0$, so $x^2 + 1 \geq 2x$, and divide by $x > 0$ to get $x + \frac{1}{x} \geq 2$.

why: Every step is reversible here - the division by x is safe because $x > 0$ - so the reversal is valid.

4 So the best description is: it assumes what it is proving, and would be valid reversed.

why: Both halves matter: the flaw, and the fact that it is repairable.

Where it goes wrong

- Calling the argument correct because each line follows from the previous one - direction is what matters.
- Calling the claim false; it is true, as the reversed proof shows.
- Thinking multiplying by x reverses the inequality - $x > 0$ is given.

Takeaway Working backwards is how you **find** a proof; to **present** one, start from a known truth and check every step reverses.

Question 5 ERR-15 · Identifying errors in proofs · spec Err1

Claim: the sum of the first n odd numbers is n^2 .

- I The first odd number is $1 = 1^2$, so the claim holds for $n = 1$.
- II Suppose the claim holds for some n , i.e. $1 + 3 + \dots + (2n - 1) = n^2$.
- III Adding the next odd number, $2n + 1$, gives $n^2 + 2n + 1 = (n + 1)^2$.
- IV So the claim holds for $n + 1$, and hence for all positive integers n .

Which of the following is correct?

Answer: A

This is a valid proof by induction: a base case, and a step showing that truth for n implies truth for $n + 1$.

Full method - how and why

A correct proof by induction. Check the base case, the inductive hypothesis, the step and the conclusion in turn.

- 1 Line I: base case $n = 1$ gives $1 = 1^2$ ✓.
why: An induction needs a case to start from.
- 2 Line II: assuming the result for some n is the **inductive hypothesis** - not circular, because it is a supposition inside the step, not a global assumption.
why: This is the point most often misidentified as an error.
- 3 Line III: the next odd number after $2n - 1$ is $2n + 1$ ✓, and $n^2 + 2n + 1 = (n + 1)^2$ ✓.
why: Check the term: odd numbers go up in twos, so $(2n - 1) + 2 = 2n + 1$.
- 4 Line IV: base case plus step gives the result for every positive integer ✓. The proof is correct.
why: $n = 1$ gives $n = 2$, which gives $n = 3$, and so on without end.

Where it goes wrong

- Calling line II circular - assuming for **one** n in order to deduce $n + 1$ is exactly how induction works.
- Claiming the next odd number is $2n - 1$; that is the n th one, already included.
- Claiming line IV proves only the $n + 1$ case - the chain from the base case gives all n .

Takeaway Induction = base case + (truth at $n \Rightarrow$ truth at $n + 1$). The inductive hypothesis is not circular reasoning.

Question 6 ERR-17 · Identifying errors in proofs · spec Err1

Claim: if $f'(a) = 0$ and $f''(a) = 0$ then f has a point of inflexion at $x = a$.

Which of the following shows the claim is false?

Answer: E

x^4 has $f'(0) = f''(0) = 0$ but a minimum at 0, not an inflexion. (x^3 does have an inflexion; x^2 has $f''(0) \neq 0$.)

Full method - how and why

To disprove a claim about a test for inflexion, produce a function satisfying the conditions but failing the conclusion.

1 Need $f'(a) = f''(a) = 0$ with **no** inflexion at a .

why: Both hypotheses must hold, and the conclusion must fail.

2 Take $f(x) = x^4$ at $a = 0$: $f'(x) = 4x^3$ so $f'(0) = 0$ ✓; $f''(x) = 12x^2$ so $f''(0) = 0$ ✓.

why: Both derivatives vanish - the hypotheses are satisfied.

3 But x^4 has a **minimum** at 0 (it is positive on both sides), not a point of inflexion.

why: An inflexion requires the concavity to change sign; $f'' = 12x^2 \geq 0$ on both sides, so it does not.

4 Check the others: x^3 does have an inflexion at 0 (so it confirms, not refutes); x^2 has $f''(0) = 2 \neq 0$; $x^3 - 3x$ at $a = 1$ has $f'(1) = 0$ but $f''(1) = 6 \neq 0$; $f(x) = x$ has $f'(0) = 1 \neq 0$.

why: Each fails one of the two hypotheses, or satisfies the conclusion.

Where it goes wrong

- Choosing x^3 , the standard **example** of the claim rather than a counterexample.
- Choosing a function where one of the two derivative conditions fails at the stated point.

Takeaway $f''(a) = 0$ does not guarantee an inflexion; the second derivative must **change sign**. x^4 at 0 is the standard counterexample.

Question 7 ERR-06 · Identifying errors in proofs · spec Err1

Claim: the function $f(x) = x^3 - 3x^2 + 3x$ is increasing for all real x .

I $f'(x) = 3x^2 - 6x + 3 = 3(x - 1)^2$.

II So $f'(x) \geq 0$ for all x , with equality only at $x = 1$.

III Since $f'(x) > 0$ for all x , f is increasing.

Which of the following is true?

Answer: A

f is (strictly) increasing because $f'(x) \geq 0$ with equality only at an isolated point, so $f(a) < f(b)$ whenever $a < b$; but line III's assertion ' $f'(x) > 0$ for all x ' is false at $x = 1$. There is no turning point at $x = 1$ (the derivative does not change sign).

Full method - how and why

A subtle one: the **claim is true** but one line asserts more than was proved. Distinguish a false conclusion from an unjustified step.

1 Check line I: $f'(x) = 3x^2 - 6x + 3 = 3(x^2 - 2x + 1) = 3(x - 1)^2$. Correct.

why: Differentiate term by term, then factorise.

2 Check line II: a square is non-negative, so $f'(x) \geq 0$, with equality only at $x = 1$. Correct.

why: The word 'only' is right - $(x - 1)^2 = 0$ just at $x = 1$.

3 Line III says ' $f'(x) > 0$ for all x ' - **false at $x = 1$ **, where $f'(1) = 0$. So line III overstates line II.

why: It asserts a strict inequality that was never established.

4 Is the claim itself true? Yes: $f' \geq 0$ everywhere with equality at a single isolated point still gives $f(a) < f(b)$ for $a < b$.

why: A derivative vanishing at one point does not create a flat stretch; there is no turning point, because f' does not change sign.

Where it goes wrong

- Concluding the claim is false because $f'(1) = 0$ - a stationary point of inflexion is not a turning point.
- Blaming line I or II, both of which are correct.
- Calling the proof completely correct, ignoring the strictness error in line III.

Takeaway $f' \geq 0$ with equality only at isolated points still means strictly increasing - but a proof must not claim strict positivity it has not shown.